The nonlocal-to-local limit for superlinear SKT cross-diffusion

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Abstract

We prove convergence from regional nonlocal diffusion to the local Shigesada–Kawasaki–Teramoto system with superlinear transition rates. For any number of species and any fixed exponent greater than one, the given nonnegative strong nonlocal solutions admit a subsequence converging strongly in space-time on every finite time interval, under the symmetric coefficient and strict diagonal-dominance assumptions of the source problem. The limit satisfies the local equation with its original initial data and homogeneous Neumann pressure flux. Both nonlinear pressures and reactions converge strongly in L^1. A power entropy gives compactness of the densities. A supplementary regularized logarithmic entropy controls weighted square-root differences, which yield an integrable pressure gradient and permit first-order testing up to the boundary.

1 Introduction and main result

Cross-diffusion describes populations whose dispersal depends on the densities of other species. In a regional nonlocal model, individuals jump between points of a bounded habitat, and the jump intensity depends on the population at the departure and arrival points. When the jump length tends to zero, one expects a local diffusion equation with no flux through the boundary. Establishing this limit requires estimates uniform in the jump length; the strong bounds for an individual nonlocal equation do not suffice.

For two species with linear transition rates, Galiano and Velasco [4, Theorem 1] proved such a localization result. Hirvonen, Jüngel and Pollino [5, Theorems 1 and 2] proved global existence and uniqueness of strong solutions, together with entropy estimates, for a multispecies model with superlinear transition rates. The localization question for those rates is stated in [5, Section 1.1] and in AIM problem 469 [1]. Existence theorems for local nonlinear cross-diffusion systems, such as [3, Theorem 2], do not identify limits of the prescribed regional jump solutions. Moreover, density compactness alone does not identify a pressure gradient or the first-order weak boundary condition; the distinction from second-order testing is relevant here [3, Remark 16].

We prove the localization result with all dimensions, species and exponents in the source problem. The coefficient, kernel and initial-data hypotheses below are exactly those of that problem. In addition to strong density convergence, we obtain an integrable weak gradient of the actual nonlinear pressure. This gives the homogeneous Neumann condition with unrestricted boundary values of the test function.

1.1 The equations and assumptions

Let $d \geq 1$, let $\Omega \subset \mathbb{R}^d$ be a nonempty smooth bounded connected domain, let $n \geq 2$, and fix $s > 1$. For $u = (u_1, \ldots, u_n) \geq 0$, set

$$P_i(u) = u_i \left(a_{i0} + \sum_{j=1}^{n} a_{ij} u_j^s \right), \qquad R_i(u) = u_i \left(b_{i0} - \sum_{j=1}^{n} b_{ij} u_j \right). \tag{1}$$

Assume $a_{i0} > 0$, $b_{i0}, b_{ij} \geq 0$, and $a_{ij} = a_{ji} \geq 0$, with

$$a_{ii} > C(s) \sum_{j \neq i} a_{ij}, \qquad C(s) = \begin{cases} \dfrac{s^2}{2(s+1)(s-1)}, & 1 < s < 2, \\ \dfrac{4(s-1)^2 + 1}{4(s+1)(s-1)}, & s \geq 2. \end{cases} \tag{2}$$

Let $J \in C_c^\infty(\mathbb{R}^d)$ be nonnegative, radial, and positive near zero, with normalization

$$\frac{1}{2} \int_{\mathbb{R}^d} J(z) z_1^2 \, \mathrm{d}z = 1.$$

Thus, by radiality,

$$\int_{\mathbb{R}^d} J(z) z_k z_l \, \mathrm{d}z = 2\delta_{kl}, \qquad M_2 := \int_{\mathbb{R}^d} J(z) |z|^2 \, \mathrm{d}z = 2d. \tag{3}$$

Write $J_\varepsilon(h) = \varepsilon^{-d} J(h/\varepsilon)$ and fix $0 \leq u_i^0 \in L^\infty(\Omega) \cap BV(\Omega)$. We consider the given global nonnegative strong solutions of

$$\partial_t u_i^\varepsilon(t,x) = \varepsilon^{-2} \int_\Omega J_\varepsilon(x-y) \left[P_i(u^\varepsilon(t,y)) - P_i(u^\varepsilon(t,x)) \right] \mathrm{d}y + R_i(u^\varepsilon(t,x)), \quad u_i^\varepsilon(0) = u_i^0. \tag{4}$$

The strong sense is that, on each finite interval, the solution belongs to $W^{1,\infty}(0,T; L^\infty(\Omega))$ and $C([0,T]; L^\infty(\Omega) \cap BV(\Omega))$, and satisfies (4) almost everywhere. This is the strong-solution setting of [5, Theorem 1]. No bound in these spaces uniform in ε is assumed.

Theorem 1.1. *Under these assumptions, every sequence $\varepsilon \to 0$ has one subsequence and a global nonnegative limit u such that, for every $T < \infty$, writing $Q_T = (0,T) \times \Omega$ and*

$$r = \frac{4s}{3s+1} > 1, \qquad r' = \frac{4s}{s-1},$$

we have

$$u_i^\varepsilon \longrightarrow u_i \quad \text{strongly in } L^p(Q_T) \quad (1 \leq p < 2s), \tag{5}$$

$$P_i(u^\varepsilon) \longrightarrow P_i(u), \quad R_i(u^\varepsilon) \longrightarrow R_i(u) \quad \text{strongly in } L^1(Q_T), \tag{6}$$

$$u_i \in L^\infty(0,T; L^s(\Omega)) \cap L^{2s}(Q_T) \cap C_w([0,T]; L^s(\Omega)),$$

$$u_i^s, \sqrt{u_i} \in L^2(0,T; H^1(\Omega)), \qquad P_i(u) \in L^r(0,T; W^{1,r}(\Omega)), \tag{7}$$

$$R_i(u) \in L^s(Q_T), \qquad \partial_t u_i \in L^r(0,T; (W^{1,r'}(\Omega))').$$

Here C_w denotes continuity for the weak L^s topology. The limit has $u_i(0) = u_i^0$ weakly in $L^s(\Omega)$ and, for every $\phi \in C^1([0,T] \times \overline{\Omega})$ with $\phi(T) = 0$,

$$-\int_{Q_T} u_i \partial_t \phi - \int_\Omega u_i^0 \phi(0) + \int_{Q_T} \nabla P_i(u) \cdot \nabla \phi = \int_{Q_T} R_i(u) \phi. \tag{8}$$

No restriction is imposed on the boundary values of ϕ.

Equation (8) is the first-order weak formulation of $\partial_t u_i = \Delta P_i(u) + R_i(u)$ with homogeneous Neumann pressure flux. It does not assert a classical normal trace. The subsequence is chosen once for all finite horizons. We do not prove uniqueness of the local weak solution, a convergence rate, convergence of the whole family, or the endpoint $s = 1$.

The proof has two complementary entropy estimates. The power entropy controls differences of u_i^s and yields higher integrability. To estimate differences of the products $u_i u_j^s$ in the pressure, we also need information near vacuum. A regularized logarithmic entropy supplies the corresponding weighted square-root differences. Their combination gives a pressure difference quotient in L^r, with $r > 1$. A nonlinear compactness argument then gives strong convergence, and the pressure quotient identifies the weak gradient and allows testing through the boundary. Section 2 establishes the estimates; Section 3 proves compactness and the limit passage.

2 Uniform entropy estimates

All constants below may depend on the fixed domain, coefficients, n, s, J, u^0 and T, but never on ε or on a removed regularization parameter. For a spatial function w, define the regional quadratic form

$$D_\varepsilon(w) = \iint_{\Omega^2} \varepsilon^{-d-2} J((x-y)/\varepsilon)|w(y) - w(x)|^2 \,\mathrm{d}x\,\mathrm{d}y. \tag{9}$$

Its form domain is $L^2(\Omega)$, and we set $D_\varepsilon(w) = +\infty$ outside that space. Every fixed-ε solution slice used below is bounded, so its powers are in this form domain. No uniform L^2 bound is implicit in the definition.

2.1 Power entropy and spatial integrability

Set $S_i = \sum_{j\neq i} a_{ij}$ and $k_i = (s+1)a_{ii} - (s-1)S_i$. Since $(s+1)C(s) > s-1$, condition (2) implies $k_i > 0$. For $1 < s < 2$ this follows from $s^2 > 2(s-1)^2$; for $s \geq 2$ the difference between $C(s)$ and $(s-1)/(s+1)$ is $1/[4(s+1)(s-1)]$.

Lemma 2.1. *For nonnegative vectors X, Y, one has*

$$\sum_i (X_i^{s-1} - Y_i^{s-1})(P_i(X) - P_i(Y)) \geq \frac{4(s-1)}{s^2} \sum_i a_{i0} |X_i^{s/2} - Y_i^{s/2}|^2 + \frac{s-1}{s^2} \sum_i k_i |X_i^s - Y_i^s|^2. \tag{10}$$

Consequently the solutions satisfy

$$\sup_{0\leq t\leq T} \sum_i \|u_i^\varepsilon(t)\|_s^s + \sum_i \int_0^T D_\varepsilon((u_i^\varepsilon)^s)\,\mathrm{d}t \leq C_T. \tag{11}$$

Proof. First take positive vectors. In the variables $v_i = u_i^{s-1}$, write $F(v) = P(u)$. Direct differentiation gives

$$\partial_i F_i = \frac{a_{i0} u_i^{2-s} + (s+1)a_{ii}u_i^2 + \sum_{j\neq i} a_{ij} u_i^{2-s} u_j^s}{s-1}, \qquad \partial_j F_i = \frac{s}{s-1} a_{ij} u_i u_j \quad (i \neq j).$$

This matrix is symmetric. With $z_i = u_i \xi_i$, its off-diagonal pair is bounded using

$$a_{ij}\left[(u_j/u_i)^s z_i^2 + (u_i/u_j)^s z_j^2 + 2s z_i z_j\right] \geq -(s-1)a_{ij}(z_i^2 + z_j^2).$$

Indeed, the first two terms are at least $2|z_i z_j|$, and the only adverse case is $z_i z_j < 0$. Hence

$$\xi^T DF(v)\xi \geq \frac{1}{s-1}\sum_i (a_{i0}u_i^{2-s} + k_i u_i^2)\xi_i^2. \tag{12}$$

Integrate along the straight segment from Y^{s-1} to X^{s-1}, and put $U_i(\theta) = ((1-\theta)Y_i^{s-1} + \theta X_i^{s-1})^{1/(s-1)}$. Differentiating U_i^s and $U_i^{s/2}$ and applying Cauchy–Schwarz gives

$$(X_i^{s-1} - Y_i^{s-1})^2 \int_0^1 U_i^2 \,\mathrm{d}\theta \geq \frac{(s-1)^2}{s^2}|X_i^s - Y_i^s|^2,$$

$$(X_i^{s-1} - Y_i^{s-1})^2 \int_0^1 U_i^{2-s} \,\mathrm{d}\theta \geq \frac{4(s-1)^2}{s^2}|X_i^{s/2} - Y_i^{s/2}|^2.$$

Together with (12) these prove (10). Continuity extends that inequality to nonnegative endpoints. Thus no fractional power is differentiated at vacuum.

Use $h_s(t) = (t^s - st + s - 1)/(s-1) \geq 0$ as entropy, and let $E(t) = \sum_i \int_\Omega h_s(u_i^\varepsilon(t))$. Symmetry of the kernel gives

$$E'(t) + \frac{2}{s}\sum_i a_{i0} D_\varepsilon((u_i^\varepsilon)^{s/2}) + \frac{1}{2s}\sum_i k_i D_\varepsilon((u_i^\varepsilon)^s) \leq C(1 + E(t)). \tag{13}$$

For the reaction bound, if $u_i \geq 1$, the competition term multiplied by $h_s'(u_i)$ is nonpositive, and the growth term is bounded by $C(1+u_i^s)$. If $u_i < 1$, the positive competition contribution is bounded by $C\sum_j u_j \leq C(1 + \sum_j u_j^s)$. Finally $u_i^s \leq C(1 + h_s(u_i))$.

The chain rule is legitimate for every fixed ε: the solution is a bounded Lipschitz time curve into L^∞, and the entropy is C^1 on its nonnegative range. At zero use the C^1 extension $(|t|^s - st + s - 1)/(s-1)$. The derivative of $U \mapsto \int h_s(U)$ is $V \mapsto \int h_s'(U)V$; the remainder is bounded by $|\Omega|\,\|V\|_\infty$ times the modulus of continuity of h_s' on the fixed bounded range. Thus the usual absolutely continuous scalar chain rule applies. All fixed-ε pair integrals are finite before symmetrization. Multiplication of the differential inequality by e^{-Ct} bounds E; integration of (13) then proves (11). □

We use the following precise spatial consequence of Ponce's results. It restores the mean before asserting a uniform norm bound.

Lemma 2.2. *For sufficiently small ε and every $w \in L^2(\Omega)$,*

$$\|w - \overline{w}\|_2^2 \leq C D_\varepsilon(w), \qquad \overline{w} = |\Omega|^{-1}\int_\Omega w. \tag{14}$$

If $\varepsilon_k \to 0$ and both $\|w_k\|_2$ and $D_{\varepsilon_k}(w_k)$ are bounded, then a subsequence converges strongly in $L^2(\Omega)$.

Proof. Recall the second moment $M_2 = \int J(z)|z|^2 \,\mathrm{d}z = 2d$ and set $\rho_\varepsilon(h) = |h|^2 J_\varepsilon(h)/(M_2\varepsilon^2)$. These are nonnegative radial probability kernels concentrating at zero, and

$$\iint_{\Omega^2} \frac{|w(y) - w(x)|^2}{|x-y|^2}\rho_\varepsilon(x-y)\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{M_2}D_\varepsilon(w).$$

For $d \geq 2$, (14) and the compactness assertion are [6, Theorems 1.1–1.2], at $p = 2$. Their hypotheses are a bounded Lipschitz domain, radial probability kernels concentrating at zero, and the displayed bounded norm and energy. The eventual Poincaré constant can be chosen uniformly for the present family: otherwise choose a violating sequence $\varepsilon_k \to 0$ and apply Theorem 1.1 to that very sequence with its fixed constant.

For $d = 1$, connectedness makes Ω an interval. Choose $R, j_0 > 0$ with $J \geq j_0$ on $[-R, R]$. If $R\varepsilon/2 \leq |h| \leq R\varepsilon$ and $1/2 \leq \theta \leq 1$, then

$$\rho_\varepsilon(\theta h) \geq \frac{j_0 R^2}{16 M_2 \varepsilon}.$$

The annulus has length $R\varepsilon$, so $\int \inf_{1/2\leq\theta\leq 1} \rho_\varepsilon(\theta h)\,\mathrm{d}h \geq j_0 R^3/(16M_2) > 0$. This verifies condition (8) of [6, Theorem 1.3]; the kernels here are smooth, so its Lebesgue-point representative causes no ambiguity. That theorem extends both assertions to $(0,1)$. For $\Omega = (a, a+L)$, use $W(x) = w(a+Lx)$ and $\widetilde{\rho}(h) = L\rho_\varepsilon(Lh)$. The original weighted energy is L^{-1} times the energy on $(0,1)$, and the original squared oscillation is L times that on $(0,1)$. Only the fixed factor L^2 changes the Poincaré constant. Compactness transfers by the same rescaling. □

Apply (14) to $w = (u_i^\varepsilon)^s$. Its mean is bounded by (11). Thus, for all sufficiently small ε,

$$\sum_i \|u_i^\varepsilon\|^{2s}_{L^{2s}(Q_T)} \leq C_T. \tag{15}$$

In particular $P_i(u^\varepsilon)$ is bounded in $L^{2s/(s+1)}(Q_T)$ and $R_i(u^\varepsilon)$ in $L^s(Q_T)$. Both exponents exceed one.

2.2 A logarithmic estimate at vacuum

To control the pressure gradient, the power estimate must be supplemented by differences of $\sqrt{u_i^\varepsilon}$ with weights from the other species. For time-dependent w, define

$$q_\varepsilon w(t,x,z) = \mathbf{1}_\Omega(x+\varepsilon z)\frac{w(t,x+\varepsilon z) - w(t,x)}{\varepsilon}, \qquad \mathrm{d}\mu = \mathrm{d}t\,\mathrm{d}x\,J(z)\,\mathrm{d}z.$$

The quotient is zero when the translated point leaves Ω; it uses no outside value of w. The measure μ on $(0,T)\times\Omega\times\mathbb{R}^d$ is finite and independent of ε, and $\|q_\varepsilon w\|^2_{L^2(\mu)} = \int_0^T D_\varepsilon(w(t))\,\mathrm{d}t$.

Lemma 2.3. *The solutions satisfy*

$$\sum_i \int_0^T D_\varepsilon(\sqrt{u_i^\varepsilon})\,\mathrm{d}t + \sum_{i,j} a_{ij} \int \min\{(u_j^\varepsilon(t,x))^s, (u_j^\varepsilon(t,x+\varepsilon z))^s\} \cdot |q_\varepsilon \sqrt{u_i^\varepsilon}|^2\,\mathrm{d}\mu \leq C_T. \tag{16}$$

The minimum in this formula is needed only on admitted pairs.

Proof. For $a, b, c, d \geq 0$ and $\delta > 0$, put $L_\delta = \log(a+\delta) - \log(b+\delta)$. Then

$$L_\delta(ac - bd) \geq \min(c,d) L_\delta(a-b) - |a-b||c-d|. \tag{17}$$

Indeed, assume $a \geq b$ by interchanging endpoints. If $c \geq d$, write $ac - bd = d(a-b) + a(c-d)$. If $c < d$, write $ac - bd = c(a-b) - b(d-c)$ and use $b\log((a+\delta)/(b+\delta)) \leq a - b$. When $c = a^s, d = b^s$, the negative term in (17) can be omitted. Also, Cauchy–Schwarz on the integral of $(t+\delta)^{-1/2}$ gives

$$(a-b)L_\delta \geq 4|\sqrt{a+\delta} - \sqrt{b+\delta}|^2. \tag{18}$$

For any $M > 0$ and $0 < \delta \leq M$,

$$|a-b|^2 \leq 8M|\sqrt{a+\delta} - \sqrt{b+\delta}|^2 + M^{-2(s-1)}|a^s - b^s|^2. \tag{19}$$

If $\max(a,b) \le M$, use the square-root factorization. Otherwise, with $a = \max(a,b) > M$, one has $(a^s - b^s)/(a-b) \ge a^{s-1} \ge M^{s-1}$.

Let $A = \max_i \sum_{j \ne i} a_{ij}$ and $a_* = \min_i a_{i0} > 0$. For vectors X, Y, write Δ for the difference of their endpoint values. Summing (17)–(18), using symmetry, and applying Young's inequality to the off-diagonal errors yields

$$\sum_i \Delta \log(X_i+\delta)\Delta P_i \ge 4a_* \sum_i |\Delta\sqrt{X_i+\delta}|^2 + 4\sum_{i,j} a_{ij} \min(X_j^s, Y_j^s)|\Delta\sqrt{X_i+\delta}|^2 - \frac{A}{2}\sum_i (|\Delta X_i|^2 + |\Delta X_i^s|^2).$$

For $A > 0$ fix $M \le \min(1, a_*/(2A))$ once and for all. Inserting (19) costs at most $4AM \le 2a_*$ of the first coefficient. Consequently, for $0 < \delta \le M$,

$$\sum_i \Delta \log(X_i+\delta)\Delta P_i \ge 2a_* \sum_i |\Delta\sqrt{X_i+\delta}|^2 + 4\sum_{i,j} a_{ij} \min(X_j^s, Y_j^s)|\Delta\sqrt{X_i+\delta}|^2 - C\sum_i |\Delta X_i^s|^2, \tag{20}$$

where $C = (A/2)(1 + M^{-2(s-1)})$. If $A = 0$, take $M = 1, C = 0$.

Test the fixed-ε equations with $\log(u_i^\varepsilon + \delta)$, the derivative of $B_\delta(t) = (t+\delta)\log(t+\delta) - (t+\delta) + 1 \ge 0$. The chain rule follows as in Lemma 2.1, using a C^1 extension to $t > -\delta/2$ at fixed δ. Uniformly for $\delta \le 1$,

$$\sum_i R_i(u)\log(u_i+\delta) \le C(1 + \sum_i u_i^s). \tag{21}$$

For $u_i + \delta \ge 1$ discard competition and use $u_i \log(u_i+1) \le C_s(1+u_i^s)$. In the other case discard growth and use $u_i|\log(u_i+\delta)| \le 1/e$, leaving a bound by $C\sum_j u_j$. The initial B_δ integral is uniformly bounded. Integrate the entropy identity, whose diffusion factor is $1/2$, and apply (20), (21) and (11). The negative power dissipation has already been bounded. All remaining dissipation terms are nonnegative. Fatou's lemma as $\delta \downarrow 0$ at fixed ε proves (16). No positive density floor or test with an unregularized logarithm at zero is used. □

3 Compactness and the local equation

3.1 Pressure differences and time derivatives

Lemma 3.1. *For $r = 4s/(3s+1)$ and $r' = 4s/(s-1)$,*

$$\|q_\varepsilon P_i(u^\varepsilon)\|_{L^r(\mu)} \le C_T, \qquad \|\partial_t u_i^\varepsilon\|_{L^r(0,T;(W^{1,r'}(\Omega))')} \le C_T. \tag{22}$$

Proof. For $a, b, c, d \ge 0$,

$$|ac - bd| \le \min(c,d)|a-b| + \max(a,b)|c-d|. \tag{23}$$

For the product $u_i^\varepsilon (u_j^\varepsilon)^s$, take a, b to be its i-species endpoint values and c, d the sth powers of its j-species values. The first term divided by ε factors, on admitted pairs, as

$$[\sqrt{\min(c,d)}(\sqrt{a} + \sqrt{b})]\,[\sqrt{\min(c,d)}\,|q_\varepsilon\sqrt{u_i^\varepsilon}|].$$

For $a_{ij} > 0$, the second bracket is bounded in $L^2(\mu)$ by (16). The first is bounded in $L^{4s/(s+1)}(\mu)$: its two factors are in L^4 and L^{4s} by (15). Endpoint evaluations preserve these

norms, by translation of x and the finite mass of J. Hölder's inequality gives their product in L^r, since $1/r = 1/2 + (s+1)/(4s)$. The second term of (23), divided by ε, is bounded in $L^{2s/(s+1)}$ by the L^{2s} norm of the endpoint maximum and the L^2 norm of $q_\varepsilon (u_j^\varepsilon)^s$. The linear pressure term is bounded in $L^{4s/(2s+1)}$ by factoring its difference through the square root. Both exponents exceed r, and μ is finite. Terms with $a_{ij} = 0$ require no weighted estimate. This proves the first bound.

For $\psi \in W^{1,r'}(\Omega)$ the extension theorem on a smooth domain [2, Theorem 9.7] and integration along segments give

$$\|q_\varepsilon \psi\|_{L^{r'}(\Omega \times J(z)\,\mathrm{d}z)} \le C \|\psi\|_{W^{1,r'}(\Omega)}. \tag{24}$$

Indeed, for an extension $E\psi$ the quotient equals the integral of $z \cdot \nabla E\psi(x + \theta\varepsilon z)$ over $0 \le \theta \le 1$ on admitted pairs. Translation and bounded kernel support give the estimate. Regional symmetry gives

$$\langle \partial_t u_i^\varepsilon, \psi \rangle = -\frac{1}{2} \int_{\Omega \times \mathbb{R}^d} J(z) q_\varepsilon P_i(u^\varepsilon) q_\varepsilon \psi \,\mathrm{d}x \,\mathrm{d}z + \int_\Omega R_i(u^\varepsilon)\psi. \tag{25}$$

Use (24), the first bound of (22), and the reaction bound in $L^s(Q_T)$. Since $r < s$ and $r' > s'$, the reaction functional is bounded in the required dual space and time exponent. □

3.2 Nonlinear space-time compactness

Fix the negative space $Y = (H_0^m(\Omega))'$ with $m = \lceil d/2 \rceil + 3$. Sobolev embedding gives $H_0^m(\Omega) \hookrightarrow W^{1,r'}(\Omega) \cap L^\infty(\Omega)$ [2, Corollaries 9.13 and 9.15]. Thus $L^1(\Omega)$ embeds continuously and injectively in Y; injectivity follows by testing against $C_c^\infty(\Omega)$. Lemma 3.1 bounds the time derivative in $L^1(0,T;Y)$.

Lemma 3.2. *For each $T < \infty$, the family along any vanishing sequence of ε is relatively compact in $L^1(Q_T)^n$.*

Proof. Consider one species. Choose C_0 bounding its spatial L^s norm at all times by (11). If $\varepsilon_k \to 0$, $a_k \ge 0$, $\|a_k\|_s \le C_0$, and $D_{\varepsilon_k}(a_k^s) \le M$, then Lemma 2.2 and the bounded mean imply compactness of a_k^s in L^2. The roots are compact in L^{2s}, since $|a-b|^s \le |a^s - b^s|$ for nonnegative a, b. In particular they are compact in L^1.

It follows that, for every $\eta > 0$, there are $\varepsilon_\eta > 0$ and $C_\eta < \infty$ such that, for $\varepsilon < \varepsilon_\eta$ and $\|a\|_s, \|b\|_s \le C_0$,

$$\|a - b\|_1 \le \eta\big[1 + D_\varepsilon(a^s) + D_\varepsilon(b^s)\big] + C_\eta \|a - b\|_Y. \tag{26}$$

Here $a, b \ge 0$, and infinite energies make the assertion vacuous. Otherwise, failure would supply $\varepsilon_k < 1/k$ violating the estimate with $C_\eta = k$. The left side has a fixed upper bound, so both energies are bounded and $\|a_k - b_k\|_Y \to 0$. By the compactness just proved, subsequences converge in L^1. Injectivity of $L^1 \hookrightarrow Y$ makes their limits equal. This contradicts the strict lower bound η for their L^1 distance.

Apply (26) to $u_i^\varepsilon(t+h), u_i^\varepsilon(t)$ and integrate. The integrated Y time shift is at most $h\,\|\partial_t u_i^\varepsilon\|_{L^1(0,T;Y)}$, so

$$\limsup_{\varepsilon \to 0} \int_0^{T-h} \|u_i^\varepsilon(t+h) - u_i^\varepsilon(t)\|_1 \,\mathrm{d}t \le \eta(T + 2C_T) + C_\eta C_T h. \tag{27}$$

First send $h \to 0$, then $\eta \to 0$.

For spatial translations, extend each density by zero outside Ω. On slices with $D_\varepsilon((u_i^\varepsilon)^s) \le M$, the spatial compactness above gives an asymptotically uniform L^1 translation modulus.

To see this, a failure would give $\varepsilon_k \to 0$, shifts $h_k \to 0$ and such slices converging in L^1 but with a nonvanishing translation defect. This contradicts translation continuity of the fixed zero-extended limit. The complementary times have measure at most C_T/M by (11). Their translation contribution is at most $2 \sup_t \|u_i^\varepsilon(t)\|_1 C_T/M$. Send the spatial shift to zero on the good slices, then $M \to \infty$.

After zero extension in time, the endpoint strips contribute at most $C_T h$. All extensions have common bounded support in $\mathbb{R}^{d+1}$. For any fixed vanishing sequence, finitely many early members have the ordinary translation continuity of a single L^1 function. Combining them with the eventual estimates and the triangle inequality gives uniform space-time translation continuity. The Fréchet–Kolmogorov theorem [2, Theorem 4.26 and Lemma 4.3] proves the assertion. □

After extraction, $u_i^\varepsilon \to u_i$ strongly in $L^1(Q_T)$ and almost everywhere. Fatou's lemma and (15), followed by interpolation, give (5). For $p = s + 1 < 2s$ the elementary estimate

$$|a^s - b^s| \le s|a - b|(a + b)^{s-1} \quad (a, b \ge 0)$$

gives strong convergence of $(u_j^\varepsilon)^s$ in $L^{p/s}$. Splitting the product difference and using Hölder with p and p/s then gives convergence of $u_i^\varepsilon (u_j^\varepsilon)^s$ in L^1. Strong L^2 convergence similarly passes all quadratic reaction products. This proves (6), without a weak-product argument. Nonnegativity and the $L_t^\infty L_x^s$ and L^{2s} bounds pass to the limit.

3.3 The pressure gradient, boundary condition and initial trace

Proof of Theorem 1.1. We first work on a fixed finite horizon. Recall the regional quotient and its fixed measure:

$$q_\varepsilon w = \mathbf{1}_\Omega(x + \varepsilon z) \frac{w(t, x + \varepsilon z) - w(t, x)}{\varepsilon}, \qquad \mathrm{d}\mu = \mathrm{d}t\, \mathrm{d}x\, J(z)\, \mathrm{d}z.$$

By (22) and reflexivity, a further subsequence satisfies $q_\varepsilon P_i(u^\varepsilon) \rightharpoonup G_i$ in $L^r(\mu)$, where $r > 1$. Take $\eta \in C_c^\infty((0, T) \times \Omega)$ and a bounded measurable function $\zeta(z)$ on $\operatorname{supp} J$. Translating the first term of the quotient, for small enough ε relative to the distance of $\operatorname{supp} \eta$ from the boundary, gives in the limit

$$\int G_i \eta \zeta \, \mathrm{d}\mu = -\int P_i(u)\, z \cdot \nabla_x \eta\, \zeta \, \mathrm{d}\mu. \tag{28}$$

Here (6) pairs with the uniformly convergent smooth test difference quotient. We have not applied a difference operator to a bare L^1 limit. Define $V_{ik}(t, x) = \frac{1}{2} \int J(z) z_k G_i(t, x, z)\, \mathrm{d}z$. Hölder's inequality and compact kernel support give $V_{ik} \in L^r(Q_T)$. Use $\zeta(z) = z_k$ in (28) and (3); then $V_{ik} = \partial_k P_i(u)$ in distributions. The pressure itself belongs to L^r by its $L^{2s/(s+1)}$ bound. Hence $P_i(u) \in L^r(0, T; W^{1,r}(\Omega))$. Returning to all product tests in (28), distributional separation and Fubini give

$$G_i(t, x, z) = z \cdot \nabla P_i(u)(t, x) \quad \mu\text{-almost everywhere.} \tag{29}$$

The identical argument at exponent two, using the power and square-root difference bounds, proves $u_i^s, \sqrt{u_i} \in L^2(0, T; H^1(\Omega))$. Their L^2 norms follow from (15) and the mass bound, respectively.

Now let $\phi \in C^1([0, T] \times \overline{\Omega})$, $\phi(T) = 0$. Extend the spatial test by the bounded $W^{1,\infty}$ extension theorem. Its Lipschitz representative bounds $|q_\varepsilon \phi|$ by $C|z|$. At every interior

point x, for small enough ε, the quotient uses only the original C^1 function. The regional indicator tends to one almost everywhere. Dominated convergence therefore gives

$$q_\varepsilon \phi \longrightarrow z \cdot \nabla \phi \quad \text{strongly in } L^{r'}(\mu).$$

Integrate (25) in time and use (6) and (29). The diffusion pairing tends to

$$\frac{1}{2} \int J(z)(z \cdot \nabla P_i(u))(z \cdot \nabla \phi)\, \mathrm{d}z = \nabla P_i(u) \cdot \nabla \phi.$$

The initial term remains exactly $\int_\Omega u_i^0 \phi(0)$. This proves (8) for arbitrary boundary values of ϕ. Its absence of a boundary contribution is precisely the weak homogeneous Neumann pressure condition.

For completeness we identify the time representative, including at zero. Put $V = W^{1,r'}(\Omega)$ and define

$$F_i(t)(\psi) = -\int_\Omega \nabla P_i(u) \cdot \nabla \psi + \int_\Omega R_i(u)\psi.$$

Its V' norm is bounded by an $L^r(0,T)$ function. Smooth functions up to the boundary are dense in V: extend to $\mathbb{R}^d$, mollify, use translation continuity for the function and its weak derivatives, then restrict to Ω. This uses the extension theorem and translation lemma already cited, and does not assert density of interior compact tests in the full boundary space. Thus (8) shows that u_i agrees almost everywhere with

$$U_i(t) = u_i^0 + \int_0^t F_i(\tau)\, \mathrm{d}\tau \quad \text{in } V'.$$

This representative is continuous in V'. At an exceptional time, approach by ordinary times, use the uniform L^s bound and reflexivity to obtain a weak L^s limit. Continuity in V' identifies it uniquely with $U_i(t)$, so the bound holds at every time. Pairings with arbitrary $L^{s'}$ tests are uniform limits of continuous pairings with smooth tests, by their density in $L^{s'}$ and the uniform L^s bound. Therefore $u_i \in C_w([0,T]; L^s)$, $u_i(0) = u_i^0$ weakly in L^s, and its time derivative has the asserted $L^r(0,T;V')$ bound.

Finally, start with any vanishing sequence and perform nested extractions on $T = 1, 2, 3, \ldots$. The diagonal subsequence converges on every finite horizon. Its strong L^1 limits agree on overlaps, defining one global u. All estimates and weak identities restrict to any finite T. This proves the theorem. □

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AI-use disclosure

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